

Major Project Report on

Intent Classification of Nepali Sentence using BiLSTM



in partial fulfillment for the award of the degree of
Bachelor of Computer Engineering

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March, 2026

DECLARATION

We hereby declare that this study entitled **Intent Classification of Nepali Sentence using BiLSTM** is based on our original research work. Related works on the topic by other researchers have been duly acknowledged. We owe all the liabilities relating to the accuracy and authenticity of the data and any other information included here under.

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RECOMMENDATION

This is to certify that this project entitled **Intent Classification of Nepali Sentence using BiLSTM** prepared and submitted by **Dibas Timilsena, Kanchan Thapa, Manita Joshi, Sangita Bhattarai** in partial fulfillment of the requirements of the degree of Bachelor in Computer Engineering awarded by Pokhara University, has been completed under my supervision. I recommend the same for acceptance by Pokhara University.

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ABSTRACT

Intent classification in Nepali, a low-resource language, presents significant challenges due to the limited availability of annotated datasets and linguistic complexities such as code-mixing and regional variations. This study focuses on the multi-class intent classification of hate and non-hate speech in Nepali social media text. Hate speech is further categorized into sub-labels such as blaming, dehumanizing, and mocking, while non-hate speech is categorized into informative, neutral, and supportive intents. Motivated by the increasing online toxicity in Nepal’s digital platforms, the primary objective of this research is to develop an effective classification system for identifying and categorizing hate speech. The proposed methodology utilizes a Bidirectional Long Short-Term Memory (BiLSTM) network to capture contextual and sequential dependencies in Nepali text. Data preprocessing includes text cleaning, stop-word removal, and vectorization using dense word embeddings. The main classification model achieves an accuracy of 79%, demonstrating balanced performance between hate and non-hate classes. For the hate sub-label classification task, the model achieves an accuracy of 69%, while the model for non-hate sub-label classification achieves an accuracy of 77%. Overall, the experimental results indicate that the proposed hierarchical classification approach is effective for Nepali hate speech detection and fine-grained intent categorization. The system demonstrates stable performance at the primary classification level and reasonable discrimination among sub-labels, highlighting its potential application in real-world Nepali language content moderation. However, further improvements are required to address challenges related to ambiguous and minority sub-classes.

Keywords: Hate speech, intent, Bi-LSTM, morphology, classification, model, hierarchy

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LIST OF ABBREVIATIONS

Bi-LSTM	Bidirectional Long Short-Term Memory
HAN	Hierarchical Attention Network
LSTM	Long Short-Term Memory
LLM	Large Language Model
MuRIL	Multilingual Representations for Indian Languages
NER	Named Entity Recognition
NFC	Normalization Form Canonical Composition
NLU	Natural Language Understanding
NLP	Natural Language Processing
SVM	Support Vector Machine

Chapter 1: INTRODUCTION

1.1 Background

With the rapid growth of social media platforms, online forums, and digital communication channels, user-generated text has become one of the primary means of information exchange in modern society. While these platforms encourage free expression, they have also facilitated the spread of abusive, hateful, and harmful content, posing serious social and psychological challenges[1]. Hate speech and offensive language can negatively impact individuals, communities, and social harmony, particularly in multilingual and culturally diverse societies[2]. Traditional English-centric NLP models fail catastrophically on Nepali text due to its morphological richness, code-mixing regional dialects and lack of standardized orthography. For instance, the same hate intent can appear as “kutta jasto”, “kutta jastai”, or “kuttya jasto” across users[3]. Most existing hate speech detection research has primarily focused on high-resource languages such as English, leaving low-resource languages like Nepali underexplored [4]. The linguistic complexity of Nepali, combined with the frequent use of Romanized Nepali text on social media, further increases the difficulty of automated intent detection.

Therefore, there is a growing need for language-specific intent classification systems that can accurately identify both hate-related and non-hate intents in Nepali text. This project addresses this gap by developing an intent classification model capable of distinguishing between multiple intent categories, including mocking, dehumanizing, blaming, supportive, informative and neutral intents.

1.2 Statement of Problem

The increasing volume of Nepali text content on social media platforms has made manual moderation of harmful content impractical and inefficient. Existing automated systems often rely on binary classification approaches that only identify whether content is hateful or not, failing to capture the underlying intent behind the message. This limitation reduces the effectiveness of content moderation systems, as different types of hate and non-hate intents require different responses.

Moreover, the lack of large, well-annotated Nepali datasets for intent classification poses a significant challenge for developing robust NLP models. The frequent use of Romanized Nepali, informal spelling variations, and context-dependent expressions further

complicates accurate classification. As a result, current models struggle with misclassification, particularly between semantically similar intents such as blaming and supportive or neutral categories.

Failure to accurately identify intent can lead to either over-censorship of benign content or under-detection of harmful speech, both of which can have serious social consequences. Therefore, there is a critical need for an automated intent classification system tailored to Nepali language data that can effectively distinguish between fine-grained hate and non-hate intents.

1.3 Objective

- To create a labeled dataset of Nepali text with multiple intent categories and develop a BiLSTM-based model for accurate intent classification.

1.4 Scope

This project focuses on intent classification of Nepali text data collected from social media and online platforms. The scope includes classification of text into predefined intent categories such as mocking, dehumanizing, blaming, supportive, informative and neutral. Both hate and non-hate intents are considered within the system. The project is limited to textual data and does not include multimodal inputs such as images, audio, or videos. Additionally, the system is designed for offline analysis and research purposes. The performance of the model is dependent on the quality and size of the annotated dataset used for training and evaluation.

1.5 Applications

- Using the labeled Nepali dataset to train AI systems for understanding user intentions.
- Employing the BiLSTM model to classify Nepali text accurately into multiple intents.
- Developing intelligent Nepali chatbots that respond based on detected user intent.
- Enhancing customer support and virtual assistant systems for Nepali-speaking users.
- Enabling intent-based information retrieval and automated response generation in Nepali.

Chapter 2: LITERATURE REVIEW

Several studies have explored the detection of offensive content such as racist, sexist, and abusive language in Nepali social media. One prominent work introduced a manually annotated dataset comprising 7,462 social media entries written in both Devanagari and Romanized Nepali scripts [5]. The authors proposed language-specific preprocessing techniques, including Dirghikaran handling and Romanization, Normalization, to address orthographic variations. Their findings revealed that traditional machine learning models, particularly Random Forest, outperformed Multilingual BERT due to the limited representation of Nepali in pretrained transformer models. This result emphasized the importance of lightweight models and tailored preprocessing for low-resource languages.

Building on classical machine learning approaches, another study experimented with Support Vector Machines (SVM), Random Forest, and LSTM-based architectures for hate speech detection[6]. The results showed that SVM combined with Word2Vec embeddings achieved superior performance for English hate speech classification, while Nepali text was mainly utilized for sentiment and topic analysis. Notably, the study demonstrated a real-world application by implementing a hate speech detection system as a Chrome extension using a React frontend and Django backend, highlighting the practical feasibility of NLP-based moderation tools.

Recent research has increasingly focused on transformer-based models for Devanagari script languages. One such study employed an ensemble of XLM-RoBERTa, MuRIL, and IndicBERT to distinguish hate speech from sensitive but non-hateful content[7]. To address severe class imbalance, the authors applied backtranslation-based data augmentation techniques, resulting in high recall and competitive F1-scores in the CHIP-SAL@COLING 2025 shared task. This work demonstrated the effectiveness of model ensembling and data augmentation in low-resource hate speech detection.

Further evaluation of transformer architectures was conducted under the NLU of Devanagari Script Languages 2025 initiative, where tasks such as language identification and hate speech targeting were analyzed across multiple languages [8]. The study reported that transformer models consistently outperformed classical approaches, with a Twitter-trained XLM-RoBERTa achieving 99.5% accuracy in language identification and an F1-score of 72.5% for hate speech detection, reinforcing the dominance of pretrained transformers in multilingual NLU tasks.

Beyond Nepali, research on code-mixed languages has contributed valuable methodological insights. A study on Tamil code-mixed offensive language detection proposed a Phrase-Based Explainable Hierarchical Attention Network (HAN) that utilized phrase-level embeddings and Focal Loss to manage class imbalance[9]. A key contribution of this work was its emphasis on explainability, where attention weights were used to highlight offensive phrases, offering interpretability beyond traditional black-box classifiers.

In the Nepali context, the introduction of the NepX-Hate benchmark marked a significant milestone[10]. This dataset consists of 10,621 tweets annotated across multiple hate dimensions, including casteism, racism, and religious intolerance. By applying multitask learning with IndicBERTv2, the study achieved an improved F1-score of 0.752. Additionally, SHAP-based interpretability techniques were used to identify tokens contributing to toxic predictions, enhancing model transparency.

Related work in intent classification has also influenced hate speech research. A study on Nepali factoid question classification developed a dataset of 5,007 questions categorized into six intent classes[11]. Using an SVM with an RBF kernel and a specialized preprocessing pipeline that preserved Nepali wh-words, the model achieved a weighted F1-score of 0.74. This work demonstrated the importance of intent-level understanding and language-aware preprocessing for Nepali NLP tasks.

Transfer learning approaches have further advanced Nepali offensive language detection. One study utilized the NepSA dataset of YouTube comments and combined Named Entity Recognition with sentiment analysis[12]. The findings indicated that language-specific models such as DistilBERT-base-nepali outperformed multilingual counterparts, achieving a sentiment classification accuracy of 72.24%, thereby underscoring the value of Nepali-focused pretrained models.

The role of Large Language Models (LLMs) in multilingual hate speech detection has also been explored. A comparative study evaluated traditional machine learning, deep learning, and LLM-based approaches across English, Spanish, and Urdu datasets[13]. The results showed that GPT-3.5 Turbo significantly outperformed transformer-based transfer learning models such as RoBERTa and XLM-R, particularly in joint multilingual settings, highlighting the superior generalization capabilities of LLMs.

To mitigate data sparsity across languages, another study proposed a multi-channel BERT-based framework that integrates English, Chinese, and multilingual BERT models while incorporating translated sentences as auxiliary inputs[14]. This approach improved robustness and achieved state-of-the-art performance on Spanish, German, and Italian benchmark datasets, demonstrating the effectiveness of multilingual and translation-aware architectures.

A more fine-grained linguistic distinction was introduced in a study that separated profanity from general offensiveness in Nepali Facebook comments[15]. Using a Bi-directional LSTM with BERT embeddings on a dataset of 11,681 comments, the model achieved an accuracy of 87.8% for profanity detection. The analysis further revealed notable gender-based patterns in offensive language usage, contributing sociolinguistic insights alongside technical findings.

Finally, a comparative analysis conducted under the Paramananda@NLU of Devanagari Script Languages 2025 initiative evaluated FastText and BERT across five Devanagari language[16]. Interestingly, FastText with augmented data outperformed BERT in hate speech detection, achieving an F1-score of 0.8552 compared to 0.5763 for BERT, while BERT proved more effective for identifying complex hate targets. This study emphasized that model selection should be guided by task complexity rather than model size alone.

Overall, the existing body of NLP literature demonstrates substantial progress in offensive language and hate speech detection; however, most studies remain limited to binary classification frameworks that distinguish only between hate and non-hate content. Common challenges identified across studies include the scarcity of large-scale annotated datasets, the prevalence of code-mixed text, and the difficulty of capturing cultural and contextual nuances using standard multilingual models. Although some efforts have explored fine-grained categories such as racism or sexism, there remains a notable research gap in analyzing the underlying intent behind both hateful and non-hateful content. Addressing this limitation, the present work advances beyond simple detection by introducing a comprehensive intent classification framework. Instead of treating hate speech as a single homogeneous category, it classifies hateful content into blaming, mocking, and dehumanizing intents, while also identifying supportive, informative, and neutral intents within non-hateful content.

Chapter 3: METHODOLOGY

3.1 System Design

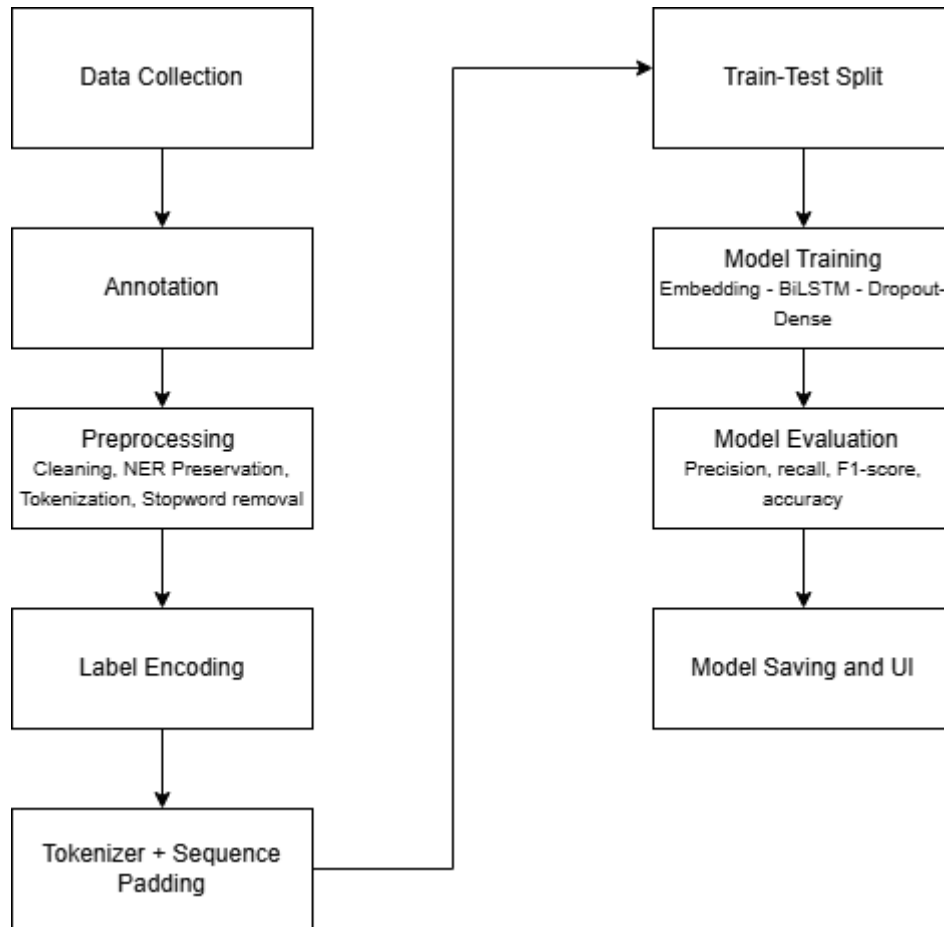


Figure 3.1 Block Diagram of System Design

The figure illustrates the overall architecture and workflow of the proposed Nepali text classification system based on the BiLSTM model. It provides a high-level overview of the process, starting from data collection and annotation, followed by text preprocessing and vectorization, label encoding, train–test splitting, model training, and final performance evaluation.

3.2 Data Collection

For this study, a comprehensive dataset was constructed through a combination of manual data collection and existing annotated resources in order to accurately reflect real-world usage of the Nepali language. Approximately 10,000 textual instances were manually collected from various social media platforms, including Facebook, YouTube, Twitter, TikTok, etc. The data collection process primarily focused on comments associated with

posts made by public figures such as celebrities and political personalities, as these sources often generate diverse and opinionated discourse. To incorporate more formal and structured language, headlines from Nepali newspapers were also included in the dataset.

In addition to the manually collected data, previously curated datasets developed by researchers at Kathmandu University (KU) were integrated into the study. The KU dataset[15] was originally created for research on *profanity and offensiveness detection in the Nepali language* and significantly contributed to enhancing the diversity and robustness of the overall dataset. The integration of both datasets resulted in a more comprehensive corpus covering a wide range of linguistic styles and expressions.

The final combined dataset contains 19,662 text in Devanagari script, Romanized Nepali, and code-switched or mixed-language content, which is commonly observed in Nepali social media communication. Each text instance, including comments and news headlines, was manually annotated to ensure labeling accuracy and contextual correctness.

3.2.1 Data Annotation

The annotation process followed a two-level labeling strategy. At the first level, all instances were classified into Hate and Non-Hate categories. At the second level, intent-based labels were assigned to provide a fine-grained classification. Hate speech instances were further categorized into Mocking, Blaming and Dehumanizing, while non-hate instances were labeled as Supportive, Informative and Neutral.

The intent categories were defined as follows. Mocking refers to the use of ridicule, sarcasm, or humiliating language aimed at insulting or belittling an individual or group. Blaming involves assigning responsibility for negative events or societal issues to a particular individual or group, often reinforcing harmful stereotypes. Dehumanizing represents language that strips individuals or groups of human qualities by comparing them to animals, objects, or diseases, thereby justifying exclusion, discrimination, or violence.

	text	main_label	sub_label
0	यो वैदेशिक हस्तक्षेप हो । अब झन छोड्नु हुन्न स...	non-hate	informative
1	१२ भाइ	non-hate	neutral
2	"जनतालाई उलु बनाउने गुरू पार्टी त एमाले नै हो,...	hate	blaming

Figure 3.2 Dataset sample

The dataset consists of 10,822 hate speech instances and 8840 non-hate instances.

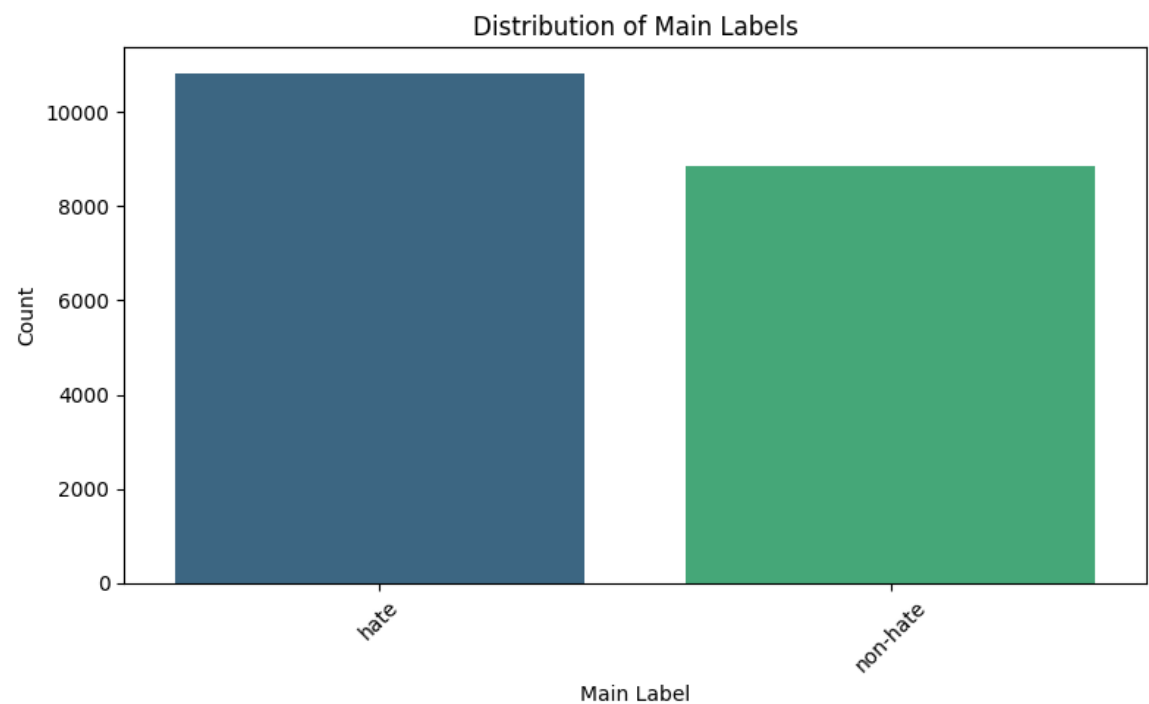


Figure 3.3 Distribution of main label

Among the annotated intent categories, mocking contains 5,144 samples, neutral includes 4723 samples, dehumanizing comprises 3430 samples, blaming contains 2249 samples, supportive includes 2089 samples, and informative consists of 2027 samples.

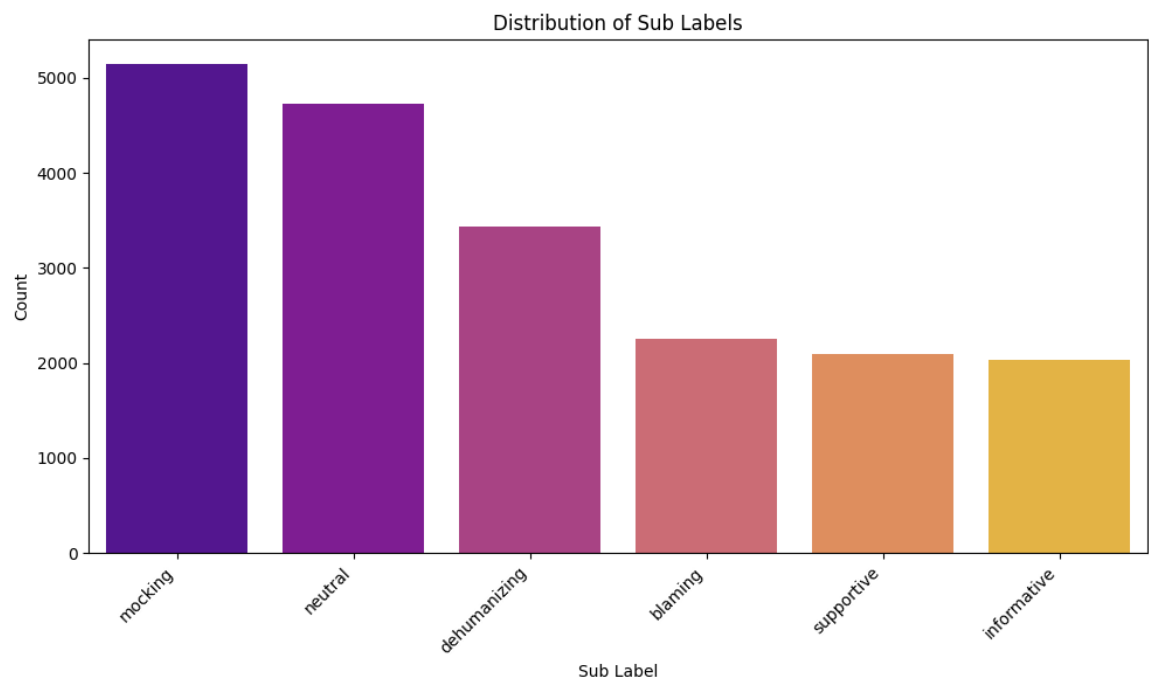


Figure 3.4 Distribution of sub labels

3.3 Data Cleaning and Preprocessing

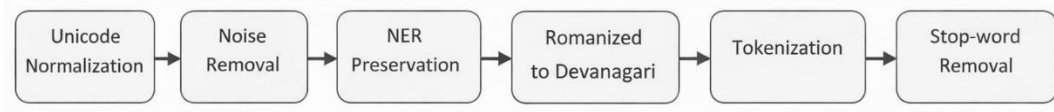


Figure 3.5 Steps of Preprocessing

Raw textual data often contains noise, inconsistencies, and irrelevant information that can negatively impact model performance. Therefore, several preprocessing steps were applied to improve the quality of the dataset.

3.3.1 Text Normalization and Cleaning

Text normalization is an essential step in Natural Language Processing, especially for languages like Nepali where text may contain inconsistent formatting.

The following operations were performed:

- Unicode normalization to standardize Nepali characters.
- Removal of unnecessary symbols and punctuation.
- Elimination of irregular spacing.
- Cleaning of unwanted characters using regular expressions.

These steps help standardize the textual data and reduce noise in the dataset.

Sample example: त एकदमै बेकार मान्छे हो 😊, तेरो सोच नै घटिया छ ।

Cleaned: त एकदमै बेकार मान्छे हो तेरो सोच नै घटिया छ

3.3.2 Named Entity Recognition for Entity Preservation

Named Entity Recognition (NER) was applied to identify important entities such as person names, locations, organizations, and geopolitical entities. These entities were preserved during further preprocessing to prevent distortion of meaningful named references. In hate speech detection tasks, named entities are often crucial because hateful expressions are frequently directed toward specific individuals, communities, or organizations. If these entities are removed during preprocessing, the contextual meaning of the sentence may be lost, which can negatively affect model performance.

To address this issue, a Named Entity Recognition–based approach was incorporated during the text cleaning process. The NER model was used to identify named entities within

the Nepali text. These identified entities were preserved while other unnecessary tokens such as stopwords, punctuation, and irrelevant characters were removed.

3.3.3 Romanized Nepali to Devanagari Transliteration

Romanized Nepali words were transliterated into Devanagari script using the Indic Transliteration library. During this process, identified named entities and pure English words were preserved in their original form to maintain semantic integrity. This step helped standardize the textual representation while avoiding unnecessary modification of proper nouns.

3.3.4 Tokenization

Tokenization is the process of breaking text into smaller linguistic units known as tokens, which typically correspond to words.

In this study, tokenization was performed using the Natural Language Toolkit (NLTK) tokenizer. Each sentence in the dataset was split into individual tokens, allowing the model to analyze the textual structure more effectively.

Sentence: त एकदमै बेकार मान्छे हो तेरो सोच नै घटिया छ

Tokenized: ['त', 'एकदमै', 'बेकार', 'मान्छे', 'हो', 'तेरो', 'सोच', 'नै', 'घटिया', 'छ']

3.3.5 Stopword Removal

To improve model efficiency and focus on features relevant to hate speech intent classification, a rigorous preprocessing phase was implemented to remove common stopwords that lack significant semantic weight. Reducing dimensionality in this manner allows the BiLSTM to prioritize intent-bearing lexical markers over structural noise. Specifically, this study adopted the linguistic framework proposed by [17], which identifies that most frequent noise in Nepali NLP originates from functional grammatical categories. Consequently, we excluded words belonging to नामयोगी, संयोजक, सर्वनाम and निपात. A predefined list of Nepali stopwords was loaded from an external text file. These stopwords were removed from the tokenized sentences to retain only meaningful words that contribute to intent classification.

After stopword removal: बेकार, मान्छे, सोच, घटिया

3.4 Label Encoding

The proposed algorithm requires numerical input; therefore, textual labels must be converted into numerical form. In this study, Label Encoding was used to transform categorical labels into numerical values using LabelEncoder class from the Scikit-learn library.

The encoding process was applied to:

- Main labels (hate, non-hate)
- Hate sub-labels
- Non-hate sub-labels

For example:

Hate - 1

Non-Hate – 0

Similarly, sub-label categories were also converted into numeric values.

3.5 Dataset Splitting

To evaluate the performance of the model, the dataset was divided into training and testing subsets using an 80:20 ratio. Eighty percent of the data was used for training the model, while the remaining twenty percent was used for testing and evaluation. Stratified sampling was applied to maintain the class distribution in both training and testing datasets.

The `train_test_split()` function from the Scikit-learn library was used for this purpose.

- Training Set: Used to train the model and learn patterns in the data.
- Testing Set: Used to evaluate the performance of the trained model on unseen data.

This separation ensures that the model's performance is assessed objectively.

3.6 Text Vectorization

Neural networks cannot process raw text directly. Therefore, textual data must be transformed into numerical vectors.

3.6.1 Word Tokenization using Keras Tokenizer

The Tokenizer class from Keras was used to build a vocabulary of the most frequent words in the dataset. Each word was assigned a unique integer index based on its frequency in the dataset.

Example:

बेकार - 1

मान्छे - 2

सोच - 3

घटिया - 4

3.6.2 Sequence Padding

Since sentences vary in length, neural networks require sequences of equal size. To solve this problem, padding was applied using the `pad_sequences()` function from Keras. Short sentences were padded with zeros, while longer sentences were truncated to maintain a fixed sequence length. The `max_length` is taken as 100 and post padding method is used. This ensures consistent input dimensions for the neural network model.

3.7 Model Development

To perform intent classification, this study utilized Bidirectional Long Short-Term Memory (BiLSTM) networks. BiLSTM works by processing Nepali text in two directions to capture the full context of every word, which is essential for identifying subtle intents. This bidirectional structure enables the model to capture contextual information from both preceding and succeeding words in a sentence, which is essential for understanding complex linguistic patterns in hate speech.

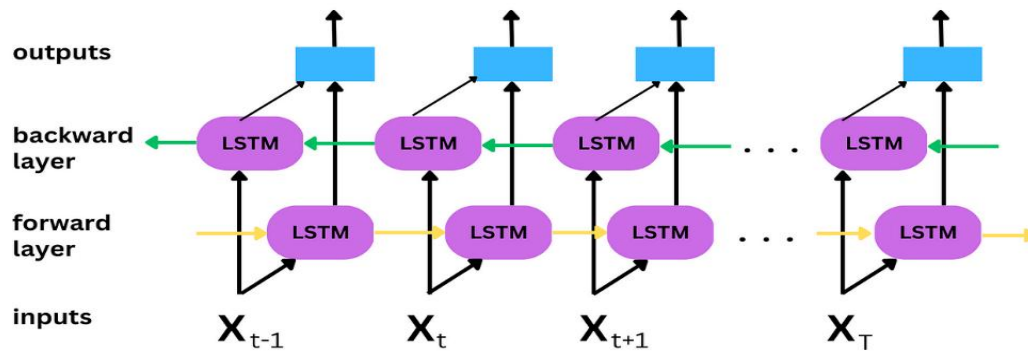


Figure 3.6 Working Structure of Bi-LSTM[18]

The architecture typically consists of the following layers:

Embedding Layer

Bidirectional LSTM Layer

Dropout Layer

Dense Output Layer

Each BiLSTM model follows the same architecture:

- **Embedding Layer:** The embedding layer transforms input text sequences into dense vector representations. These embeddings capture semantic relationships between words and help the model understand contextual similarities. The embeddings may be initialized using pre-trained word vectors or learned during training. Embedding layer helps the model understand the semantic structure of Nepali text.

बेकार - [0.12, 0.45, 0.76, 0.33]

- **Bidirectional LSTM Layer:** The core component of the model is a Bidirectional LSTM layer, which processes the input text in both forward and backward directions.

Forward direction (left $\rightarrow$ right)

Backward direction (right $\rightarrow$ left)

This enables the model to learn contextual dependencies from both preceding and succeeding words, improving its ability to detect subtle and context-dependent hate expressions.

- **Dropout Layer:** Deep learning models are prone to overfitting, especially when trained on limited datasets. Overfitting occurs when a model performs well on training data but poorly on unseen data. A dropout layer is incorporated after the BiLSTM layer to reduce overfitting. By randomly deactivating a fraction of neurons during training, the model improves generalization and becomes more robust to unseen data. If the dropout rate is 0.5 it means 50 % of neurons are randomly ignored during each training step.
- **Fully Connected (Dense) Output Layer:** The final layer in the network is the Dense (fully connected) output layer. This layer performs the final classification task. For binary classification (Hate vs Non-Hate), the dense layer typically uses the sigmoid activation function.

Dense(1, activation = 'sigmoid')

The sigmoid function outputs a value between 0 and 1, representing the probability of the input belonging to a particular class.

3.8 Hierarchical Classification Approach

The system uses a hierarchical classification strategy consisting of three separate models.

1. **Main Classifier (Hate vs. Non-Hate).**
2. **Hate Sub-classifier (Mocking vs. Blaming vs. Dehumanizing)** — only on data labelled as Hate from Level 1.
3. **Non-Hate Sub-classifier (Supportive vs. Informative vs. Neutral)** — only on data labeled as Non-Hate from Level 1.

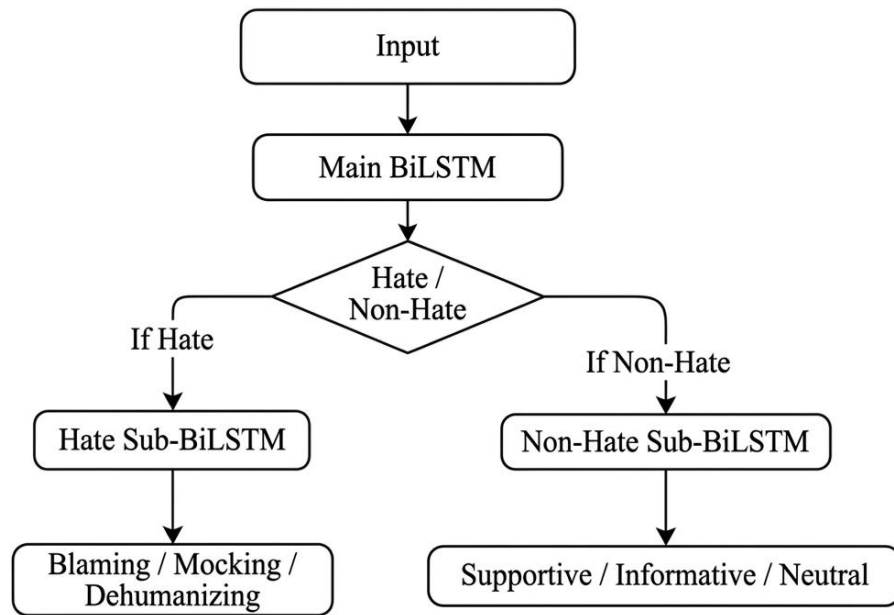


Figure 3.7 Model Hierarchy

Main Label Classification Model

The first model identifies whether the text belongs to:

- Hate
- Non-Hate

Binary classification was performed using the binary cross-entropy loss function.

Non-Hate Sub-label Classification Model

If the main model predicts non-hate, the text is passed to a second model that classifies it into:

- Informative
- Neutral
- Supportive

This is a multi-class classification problem, and the model uses sparse categorical cross-entropy as the loss function.

Hate Sub-label Classification Model

Similarly, texts predicted as hate speech are passed to another model that identifies the specific intent category such as:

- Blaming
- Dehumanizing
- Mocking

This step helps provide deeper insight into the nature of hateful content.

3.9 Handling Class Imbalance

In real-world datasets, some classes appear less frequently than others. This class imbalance can negatively affect model performance.

To address this issue, class weights were computed using the Scikit-learn library and applied during model training. This technique gives higher importance to minority classes, improving the model's ability to detect less frequent categories.

Calculated Class Weights (hate sub-labels):

Class 0 (blaming): 1.60

Class 1 (dehumanizing): 1.05

Class 2 (mocking): 0.70

3.10 Hyperparameter Optimization

Several hyperparameters were tested to improve model performance. These include:

- Embedding dimensions (64, 128, 256)
- Number of LSTM units
- Dropout rates
- Dense layer sizes
- Batch size
- Learning rate

Different configurations were evaluated, and the best-performing setup was selected based on validation accuracy.

For Hate sub label here are the best hyperparameters:

Embedding Dimension: 128.0

LSTM Units: 16.0

Dropout Rate: 0.5

Dense Layer Units: 16.0

3.11 Model Training

The model was compiled using the `compile()` function, which defines the loss function, optimizer, and evaluation metrics. The sparse categorical crossentropy loss function was used because the task involves multi-class classification with integer-encoded labels. The Adam optimizer was selected for efficient weight updates and faster convergence during training. Additionally, Early Stopping was implemented to prevent overfitting. Early stopping monitors validation loss and stops training when the performance stops improving. It saves training time and stop training at the optimal point.

`patience=10`

If validation loss does not improve for 10 consecutive epochs, the training process will stop automatically.

3.12 Model Evaluation

The performance of the trained models is evaluated using standard evaluation metrics. These metrics help measure how accurately the model classifies the text into hate and non-hate categories as well as their respective sub-labels. The main evaluation metrics used in this study include Accuracy, Precision, Recall, F1-score, and Confusion Matrix.

3.12.1 Confusion Matrix

The Confusion Matrix is the foundation table that shows exactly where the model is getting things right and where it is getting confused. The confusion matrix consists of four components:

- True Positive (TP): Hate speech correctly classified as hate speech.
- True Negative (TN): Non-hate speech correctly classified as non-hate.
- False Positive (FP): Non-hate speech incorrectly classified as hate speech.

- False Negative (FN): Hate speech incorrectly classified as non-hate speech.

In this research, the confusion matrix was used to analyze the classification performance of the main label model as well as the sub-label classification models.

3.12.2 Accuracy

Accuracy is the most common evaluation metric used in classification tasks. It measures the overall correctness of the model by calculating the proportion of correctly predicted instances out of the total number of predictions made. It is the percentage of total predictions that were correct (both Hate and Non-Hate). It shows "How often is the model right overall?"

$$\text{Accuracy} = \frac{TP+TN}{\text{Total Predictions}} \quad \text{Eq.(3.1)}$$

A higher accuracy value indicates better overall model performance. However, accuracy alone may not be sufficient when the dataset is imbalanced because it does not consider the distribution of individual classes.

3.12.3 Precision

Precision measures the proportion of correctly predicted positive observations out of all observations predicted as positive by the model. Out of all the times the model predicted "Hate", "how many were actually "hate"? "When the model flags something, how trustworthy is it?"

$$\text{Precision} = \frac{TP}{TP+FP} \quad \text{Eq.(3.2)}$$

A high precision value means that the model makes fewer false positive errors. This is particularly important when misclassifying non-hate content as hate speech could lead to unnecessary censorship or moderation.

3.12.4 Recall

Recall, also known as Sensitivity, measures the proportion of correctly predicted positive observations out of all actual positive instances in the dataset. Out of all the actual "Hate" cases that existed, how many did the model manage to catch? 'How many 'Hate' cases did we miss?'

$$\text{Recall} = \frac{TP}{TP + FN} \quad \text{Eq.(3.3)}$$

A high recall value means the model is capable of detecting most of the hate speech present in the dataset. This metric is important because failing to detect hate speech could allow harmful content to remain unmoderated.

3.12.5 F1-Score

It is the harmonic mean of Precision and Recall. It is the best metric to use if you want a single number that balances both “trustworthiness” (Precision) and “coverage” (Recall). If one is very low, the F1-score will drop significantly. This metric is particularly useful when the dataset is imbalanced because it provides a better measure of model performance than accuracy alone.

A high F1-score indicates that the model has both high precision and high recall, meaning it can correctly identify hate speech while minimizing classification errors.

3.13 Model Saving and User Interface

After training, the models were saved for future use. The saved components include:

- Main label classification model
- Hate sub-label classification model
- Non-hate sub-label classification model
- Tokenizer
- Stopword list

The saved artifacts allow the model to be reused for predictions in application which is a streamlit based user interface. UI provides an input section in which the user enters any sentence and the models predict the main label followed by sub-label.

Chapter 4: RESULT AND ANALYSIS

This chapter presents the experimental results and performance analysis of the proposed Nepali speech intent classification system.

4.1 Result

Three separate BiLSTM models were trained to classify (i) main labels, (ii) hate sub-labels, and (iii) non-hate sub-labels. The performance of each model is evaluated using standard metrics such as accuracy, precision, recall, and F1-score.

Table 4. 1Classification Result of Main Label

Class	Precision	Recall	F1-Score	Support
Hate	0.83	0.77	0.80	2,165
Non-hate	0.74	0.81	0.77	1,768
Accuracy			0.79	3,933
Macro Avg	0.79	0.79	0.79	3,933
Weighted Avg	0.79	0.79	0.79	3,933

The evaluation was conducted on a test set consisting of 3,933 sentences, where 2,165 belong to the hate class and 1,768 belong to the non-hate class.

For the hate class, the model achieved a precision of 0.83, which indicates that 83% of the sentences predicted as hate speech by the model were actually hate speech. The recall value of 0.77 shows that the model successfully identified 77% of the total hate speech instances present in the dataset. The F1-score of 0.80 represents a balanced measure between precision and recall, indicating that the model performs well in detecting hate speech while maintaining reasonable accuracy in its predictions.

For the non-hate class, the model obtained a precision of 0.74, meaning that 74% of the sentences predicted as non-hate were correctly classified. The recall value of 0.81 indicates that the model was able to correctly identify 81% of all non-hate sentences in the dataset. The F1-score of 0.77 reflects the overall balance between precision and recall for the non-hate category.

The overall accuracy of the model is 0.79, which means that the model correctly classified 79% of the total sentences in the dataset. Additionally, the macro average and weighted average F1-scores are both 0.79, indicating consistent performance across both classes. The macro average treats both classes equally, while the weighted average takes the number of samples in each class into account.

Overall, these results demonstrate that the proposed BiLSTM-based hierarchical classification model performs effectively in distinguishing between hate and non-hate Nepali text, achieving a balanced performance across both categories.

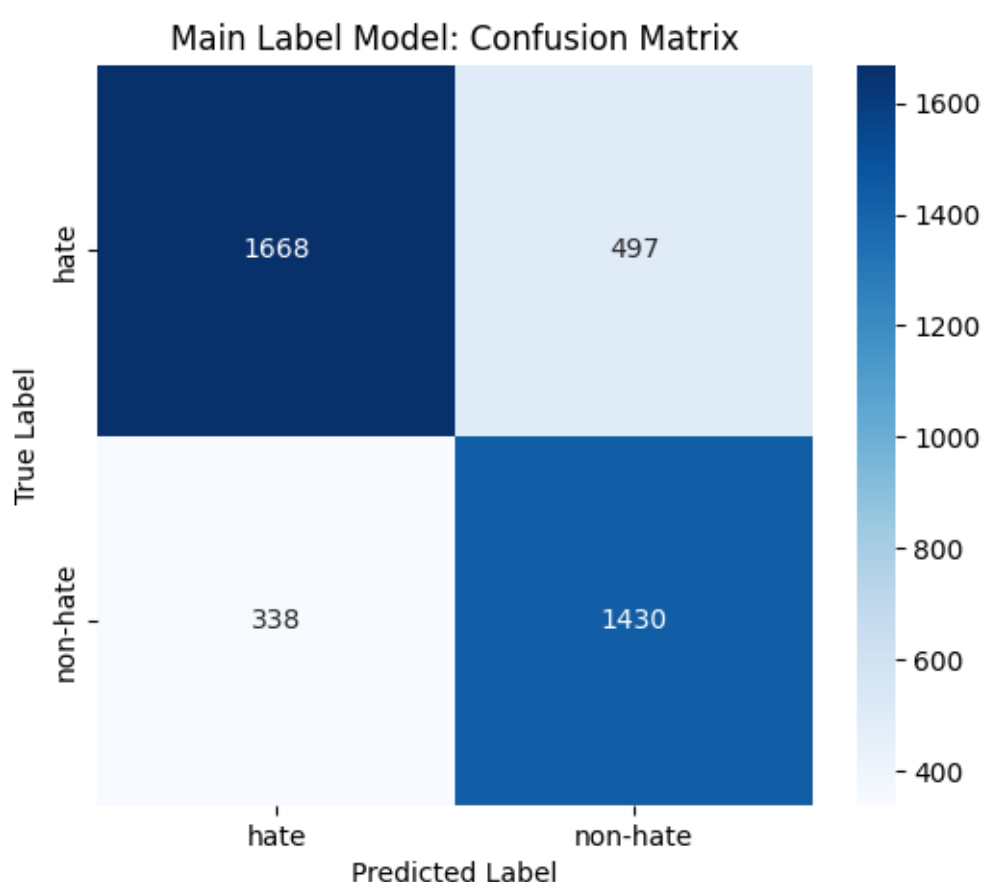


Figure 4.1 Confusion Matrix of main label

The value 1668 in the top-left cell indicates the number of sentences that were actually hate speech and were correctly predicted as hate by the model. This is called the True Positive (TP) for the hate class. The value 497 in the top-right cell represents sentences that were actually hate speech but were incorrectly predicted as non-hate. These are known as False Negatives (FN) because the model failed to detect some hate speech instances.

Similarly, the value 1430 in the bottom-right cell represents sentences that were actually non-hate and were correctly predicted as non-hate by the model. This is known as the True Negative (TN). The value 338 in the bottom-left cell shows sentences that were actually non-hate but were incorrectly classified as hate. These are called False Positives (FP), meaning the model mistakenly identified some non-hate sentences as hate speech.

Overall, the confusion matrix shows that the model correctly classified a large number of samples in both categories, with 1668 correct hate predictions and 1430 correct non-hate predictions. However, there are still some misclassifications, including 497 hate sentences predicted as non-hate and 338 non-hate sentences predicted as hate. These errors indicate areas where the model can be further improved, such as better capturing contextual or linguistic patterns in Nepali text.

Table 4. 2 Hate Sub-Label Classification Result

	Precision	Recall	F1-Score	Support
Blaming	0.70	0.48	0.57	450
Dehumanizing	0.65	0.65	0.65	686
Mocking	0.71	0.80	0.75	1029
Accuracy			0.69	2165
Macro Avg	0.69	0.64	0.66	2165
Weighted Avg	0.69	0.69	0.68	2165

For the Blaming category, the model achieved a precision of 0.70, indicating that 70% of the sentences predicted as blaming were correctly classified. The recall value of 0.48 shows that the model was able to identify only 48% of the actual blaming instances present in the dataset. This relatively lower recall suggests that the model misses a significant number of blaming-related hate speech examples. The F1-score of 0.57 reflects the balance between precision and recall for this category, indicating moderate performance. The support value of 450 represents the total number of blaming samples in the dataset.

For the Dehumanizing category, the model obtained a precision of 0.65 and a recall of 0.65, which means that the model performs consistently in both correctly predicting dehumanizing content and identifying the actual instances present in the dataset. The F1-

score of 0.65 indicates a balanced performance for this category. The dataset contains 686 dehumanizing samples, which is reflected in the support value.

For the Mocking category, the model shows comparatively better performance. It achieved a precision of 0.71, meaning that 71% of the predicted mocking instances were correct. The recall value of 0.80 indicates that the model successfully detected 80% of the actual mocking hate speech examples in the dataset. The F1-score of 0.75 demonstrates a strong balance between precision and recall, suggesting that the model is particularly effective at identifying mocking-type hate speech. The support value of 1029 indicates that mocking has the largest number of samples among the three categories.

The overall accuracy of the subclass classification model is 0.69, meaning that 69% of the hate speech samples were correctly classified into their respective subclasses. The macro average F1-score is 0.66, which represents the average performance across all classes while giving equal importance to each category. The weighted average F1-score is 0.68, which takes into account the number of samples in each class, indicating that the model performs reasonably well overall but could still be improved, particularly in detecting blaming type hatespeech.

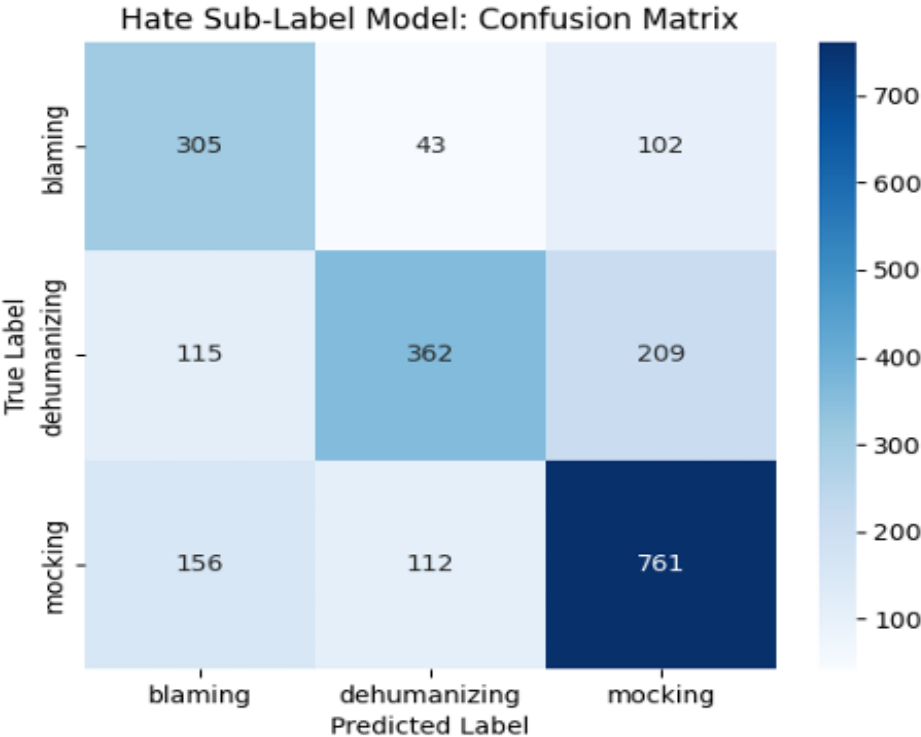


Figure 4.2 Confusion Matrix of Hate sub-label

For the Blaming class, out of all actual blaming instances, 305 sentences were correctly predicted as blaming, which represents the correct classifications for this category. However, 43 blaming sentences were incorrectly predicted as dehumanizing, and 102 blaming sentences were misclassified as mocking. This indicates that the model sometimes confuses blaming with other hate categories, particularly mocking.

For the Dehumanizing class, 362 sentences were correctly classified as dehumanizing. However, the model misclassified 115 dehumanizing sentences as blaming and 209 sentences as mocking. This suggests that the model has some difficulty distinguishing dehumanizing content from the other hate speech categories.

For the Mocking class, the model performs comparatively better. Out of all mocking instances, 761 sentences were correctly predicted as mocking. Nevertheless, 156 mocking sentences were incorrectly predicted as blaming, and 112 sentences were predicted as dehumanizing. Despite these errors, mocking has the highest number of correct predictions among the three classes.

Overall, the confusion matrix indicates that the model performs reasonably well in identifying mocking hate speech, while it shows relatively more confusion between blaming and dehumanizing categories. These misclassifications highlight areas where the model could be further improved, possibly by incorporating more training data or enhancing feature representation to better capture subtle differences between hate speech subclasses.

Table 4. 3 Non-Hate Sub-Label Classification Result

	Precision	Recall	F1-Score	Support
Informative	0.72	0.53	0.61	405
Neutral	0.76	0.90	0.82	945
Supportive	0.85	0.73	0.79	418
Accuracy			0.77	1768
Macro Avg	0.78	0.72	0.74	1768
Weighted Avg	0.77	0.77	0.77	1768

For the Informative class, the model achieved a precision of 0.72, meaning that 72% of the sentences predicted as informative were actually correct. The recall is 0.53, which indicates

that the model correctly identified only 53% of all actual informative sentences in the dataset. The F1-score is 0.61, representing the balance between precision and recall. The relatively lower recall suggests that the model misses several informative instances during prediction. The support value of 405 indicates that there were 405 informative samples in the test dataset.

For the Non-hate class, the model performs comparatively better. It achieved a precision of 0.76, meaning most predictions for this class are correct. The recall is 0.90, which shows that the model successfully identifies 90% of all actual non-hate instances. The F1-score of 0.82 indicates strong overall performance for this class. The support value is 945, which means this class has the highest number of samples among the three categories.

For the Supportive class, the model obtained a precision of 0.85, indicating that predictions for supportive content are highly accurate. The recall is 0.73, meaning the model correctly detects 73% of supportive sentences. The F1-score is 0.79, which reflects a good balance between precision and recall. The support value of 418 shows the number of supportive samples in the dataset.

Overall, the model achieved an accuracy of 0.77, meaning it correctly classified 77% of the total 1,768 samples in this non-hate subclass classification task. The macro average (0.74 F1-score) represents the average performance across all classes equally, while the weighted average (0.77 F1-score) considers the class distribution and gives more weight to classes with more samples. This indicates that the model performs reasonably well overall, with particularly strong performance in identifying the Non-hate and Supportive categories, while the Informative class remains comparatively more challenging for the model to detect accurately.

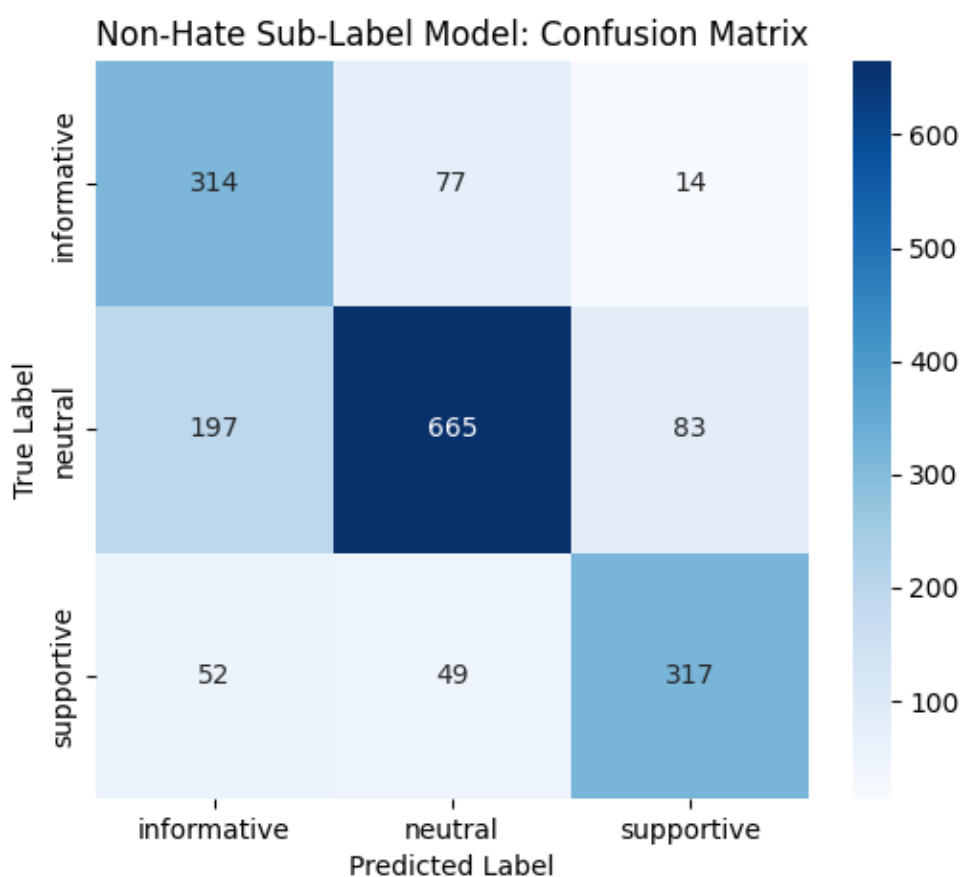


Figure 4.3 Confusion Matrix of Non-hate sub label

For the Informative class, the model correctly classified 314 sentences as informative. However, 77 informative sentences were incorrectly predicted as neutral, and 14 were predicted as supportive. This indicates that the model sometimes confuses informative content with neutral content, which may happen because both types of sentences often contain factual or descriptive information.

For the Neutral (Non-hate) class, the model achieved the highest number of correct predictions. Out of all neutral sentences, 665 were correctly classified as neutral. However, 197 neutral sentences were misclassified as informative, and 83 were misclassified as supportive. Although the model performs well in identifying neutral content, some overlap exists between neutral and informative expressions.

For the Supportive class, 317 sentences were correctly predicted as supportive. However, 52 supportive sentences were incorrectly classified as informative, and 49 were predicted as neutral. This suggests that while the model is relatively good at detecting supportive content, it sometimes confuses it with neutral or informative statements.

Overall, the confusion matrix indicates that the model performs reasonably well in identifying neutral and supportive sentences, while informative sentences are more frequently misclassified as neutral. These results highlight that the semantic similarity between informative and neutral content can make classification challenging. Further improvements could be achieved by enhancing feature representation or increasing training data to better capture the subtle differences between these non-hate speech categories.

4.2 Frequent Word Representation using Word Cloud

Word Cloud is used to visualize the most frequent words in the dataset. Words with higher frequency appear larger, while less frequent words appear smaller. This visualization helps to quickly identify commonly used words in the text data.



Figure 4.4 Word Cloud for Hate Speech



Figure 4.5 Word Cloud for Non-Hate Speech

4.3 Error Analysis

Although the proposed BiLSTM models achieved satisfactory performance in detecting hate speech intents, some misclassifications were observed during evaluation. Error analysis was conducted to understand the limitations of the model and identify possible reasons for incorrect predictions.

1. Misclassification between Hate and Non-Hate

Some sentences that contained sarcasm or indirect expressions were incorrectly classified. In certain cases, the model predicted non-hate speech as hate speech due to the presence of strong negative words.

For example, some sentences containing criticism or disagreement were incorrectly classified as hate speech because the model relied heavily on specific keywords rather than the full contextual meaning.

2. Confusion between Sub-label Categories

Errors were also observed in the classification of sub-labels. Some hate speech instances were incorrectly categorized between blaming, mocking, and dehumanizing, as these categories often share similar linguistic patterns.

Similarly, non-hate sub-labels such as informative, neutral, and supportive sometimes overlap in meaning, making it difficult for the model to distinguish between them accurately.

3. Limited Context Understanding

The model sometimes failed to capture the full semantic context of longer sentences. This limitation may occur because the model primarily focuses on word sequences and may not fully understand complex contextual relationships in Nepali text.

4. Dataset Limitations

Another reason for misclassification could be the limited size and imbalance of the dataset. Some categories contain fewer training examples, which makes it difficult for the model to learn sufficient patterns for accurate prediction.

5. Language Complexity

Nepali language often includes informal expressions, slang, and code-mixed text, which may not be well represented in the training dataset. As a result, the model sometimes struggles to correctly interpret such variations.

4.4 Discussion

The experimental results demonstrate that the proposed BiLSTM-based approach is effective for intent classification of hate speech in Nepali text.

The main label classification model achieved an accuracy of approximately 79%, indicating that the model can reasonably distinguish between hate and non-hate speech. This suggests that the deep learning architecture successfully captures important linguistic patterns related to hate speech.

For sub-label classification, the hate speech intent model achieved an accuracy of around 69%, while the non-hate intent classification model achieved approximately 77% accuracy. The slightly lower performance of the sub-label models may be attributed to the increased complexity of multi-class classification and the overlapping nature of intent categories.

Despite these challenges, the results demonstrate that the hierarchical classification approach can effectively capture different types of hate speech intent.

The use of Bidirectional LSTM allowed the model to capture contextual information from both forward and backward directions in the text sequence, which improved the model's ability to understand sentence structure.

However, further improvements could be achieved by:

- Increasing the dataset size
- Using advanced language models
- Incorporating contextual embeddings such as transformer-based models.

Overall, the findings indicate that deep learning based approaches have strong potential for automated hate speech detection in low-resource languages such as Nepali.

Chapter 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This study presented a Intent Classification System for Nepali text using a BiLSTM deep learning model. The model was trained to classify text into a main label (Hate and Non-Hate) and further into sub-label categories.

The experimental results demonstrate that the proposed BiLSTM model achieved an overall accuracy of 79% in the main label classification task. In sub-label classification, the model obtained 69% accuracy for Hate subcategories and 77% accuracy for Non-Hate subcategories. These results indicate that the model is capable of capturing contextual information in Nepali text and performing reasonably well despite the challenges associated with limited annotated datasets and linguistic complexity.

The comparatively lower performance in sub-label classification suggests that fine-grained hate categories are more difficult to distinguish due to overlapping semantics, informal language usage, spelling variations, and code-mixed Nepali-English content. Nevertheless, the system provides a strong baseline for Nepali hate speech detection and demonstrates the effectiveness of deep learning approaches for low-resource languages.

Overall, this research contributes to the development of automated content moderation tools in the Nepali language and highlights the importance of building language-specific NLP solutions for social media monitoring and online safety.

5.2 Future Work

Although the proposed system achieved promising results, several improvements can be explored in future research.

First, performance can be enhanced by increasing the size and diversity of the dataset. Collecting data from multiple social media platforms such as Facebook, Twitter (X), and YouTube would improve model generalization and robustness.

Second, advanced transformer-based language models such as BERT, mBERT, and XLM-RoBERTa can be implemented and fine-tuned for Nepali text. These models are capable of capturing deeper contextual relationships and may significantly improve sub-label classification accuracy.

Third, future research may focus on multi-class fine-grained categorization of hate speech (e.g., gender-based, religious, ethnic, political hate). This would provide more detailed analytical insights instead of simple binary classification.

Lastly, an add-on module can be designed for automatic speech detection and removal from user generated content. This add-on can be integrated with platforms such as social media websites, online forums, or chat applications to monitor text in real time.

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APPENDIX

Appendix A: UI Screenshots

Hierarchical Neural Framework for Automated Intent Analysis

 **Sequence Input**

Provide Nepali text for classification:

सबैरो बलेन बलेन मनि के गर्नु? उसको घमण्ड हामीले देखेका हौं।

Execute Inference Engine

Disclaimer: This system is an undergraduate research prototype. Due to the limited size of the training corpus and the linguistic complexity of the Nepali language, the model may exhibit marginal errors in real-world scenarios and does not guarantee absolute accuracy for all input sequences.

STATE: AGGRESSIVE INTENT DETECTED

MOCKING

Softmax Probability: 99.98%

Intent Probability Breakdown:

Blaming

Dehumanizing

Mocking

Hierarchical Neural Framework for Automated Intent Analysis

 Sequence Input

Provide Nepali text for classification:

"समस्या भए पनि मानिसले एक अर्कालाई सहयोग गरेर समाधान खोज्न सक्छन्।"

Execute Inference Engine

Disclaimer: This system is an undergraduate research prototype. Due to the limited size of the training corpus and the linguistic complexity of the Nepali language, the model may exhibit marginal errors in real-world scenarios and does not guarantee absolute accuracy for all input sequences.

STATE: NEUTRAL INTENT DETECTED

SUPPORTIVE

Softmax Probability: 100.00%

Intent Probability Breakdown:

Informative

Neutral

Supportive

Hierarchical Neural Framework for Automated Intent Analysis

 Sequence Input

Provide Nepali text for classification:

सडकमा आज ट्रफिक सामान्य थियो।

Execute Inference Engine

Disclaimer: This system is an undergraduate research prototype. Due to the limited size of the training corpus and the linguistic complexity of the Nepali language, the model may exhibit marginal errors in real-world scenarios and does not guarantee absolute accuracy for all input sequences.

STATE: NEUTRAL INTENT DETECTED

NEUTRAL

Softmax Probability: 99.99%

Intent Probability Breakdown:

Informative

Neutral

Supportive

Sequence Input

Provide Nepali text for classification:

नेपालमा प्रतिनिधि सभाको निर्वाचनपछि देशभर मतगणना भइरहेको छ। करिब ६०% मतदाता सहभागिता भएको बताइएको छ र अन्तिम परिणाम आउन केही दिन लाग्न सक्छ।

Execute Inference Engine

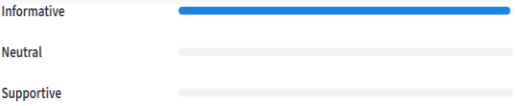
Disclaimer: This system is an undergraduate research prototype. Due to the limited size of the training corpus and the linguistic complexity of the Nepali language, the model may exhibit marginal errors in real-world scenarios and does not guarantee absolute accuracy for all input sequences.

STATE: NEUTRAL INTENT DETECTED

INFORMATIVE

Softmax Probability: 100.00%

Intent Probability Breakdown:



Sequence Input

Provide Nepali text for classification:

“राजनीतिक स्वार्थका कारण धेरै महत्वपूर्ण योजनाहरू अघि बढ्न सकेका छैनन्, जसले देशलाई पछि धकेलेको छ।”

Execute Inference Engine

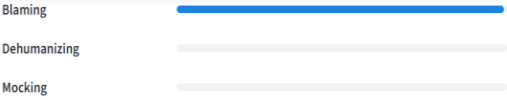
Disclaimer: This system is an undergraduate research prototype. Due to the limited size of the training corpus and the linguistic complexity of the Nepali language, the model may exhibit marginal errors in real-world scenarios and does not guarantee absolute accuracy for all input sequences.

STATE: AGGRESSIVE INTENT DETECTED

BLAMING

Softmax Probability: 100.00%

Intent Probability Breakdown:



Sequence Input

Provide Nepali text for classification:

ओती जस्ता चोरहरु लाई त देश निकाला गर्नु पर्छ ।

Press Ctrl+Enter to apply

Execute Inference Engine

Disclaimer: This system is an undergraduate research prototype. Due to the limited size of the training corpus and the linguistic complexity of the Nepali language, the model may exhibit marginal errors in real-world scenarios and does not guarantee absolute accuracy for all input sequences.

STATE: AGGRESSIVE INTENT DETECTED

DEHUMANIZING

Softmax Probability: 99.99%

Intent Probability Breakdown:

